

Placebo-Calibrated Kink Tests of Four Claimed Trend Breaks in Epoch AI Data

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estimates of record 2026-07-16 (**natex** v0.2.0); inference recalibrated 2026-10-05 (**natex** v0.3.0)

Abstract

Four claims that AI progress bent at a dated event are tested as regression kinks in calendar time on Epoch AI’s public datasets. The inference of record is the placebo-calibrated p : the rank of the declared date’s slope change among the same statistic at 46 to 49 shifted placebo dates. GPQA-Diamond scores bend at the o1-preview release and the bend is dated: the declared cutoff is the most extreme of every placebo grid at bandwidths from 365 to 730 days (calibrated $p = 0.02$). The METR 50% time-horizon slope roughly triples over the same era, but a placebo cutoff ten months earlier produces a larger statistic, so the bend cannot be attributed to the release (calibrated $p = 0.28$). The Epoch Capabilities Index shows no kink at the release (calibrated $p = 0.78$). China’s legal AI-chip stock bends down relative to US hyperscalers at the October 2023 export controls in every specification, but the quarterly series admits only seven placebo positions, so no 5% calibrated claim is possible and the result is inconclusive. Nominal HC1 inference rejects at 14 to 88% of placebo dates on these series and is not used for any verdict.

1 Introduction

Public discussion of AI progress is organised around dated events. Reasoning models are said to have changed the slope of capabilities in late 2024. Export controls are said to have bent China’s compute accumulation. Such claims are usually made by drawing a line through a scatter plot. A slope change at a known date is also a formal estimand: the regression kink design (RKD) of Card et al. [2], and for one group relative to another, the difference-in-kinks (DiK) design of Böckerman, Jysmä and Kanninen [1].

This paper tests four such claims on Epoch AI’s public data [3] with the **natex** kink module [4]. Every cutoff is externally dated and none was searched for. The question is narrow: does the series bend at the named date, more than it bends at other dates?

That last clause is the whole method. A calendar-time kink has no running-variable manipulation to test and one untestable identifying assumption, that nothing else bent the series at that date. What can be tested is whether the date is special. The same estimator is run at dozens of shifted placebo dates, and the declared date’s statistic is ranked among them. A series

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that accelerates over an era, or that is smoothly convex, rejects at many placebo dates and the declared date is not special. A series that bends once, at the date, is.

An earlier version of this paper (July 2026) used seven placebo dates per series and nominal heteroskedasticity-robust p -values. Seven placebos cannot calibrate below $p = 0.125$, and the nominal p -values are badly oversized on these series. Both problems are repaired here. The point estimates are unchanged; two of the four verdicts are not.

2 Data

All series are Epoch AI datasets (CC-BY 4.0, <https://epoch.ai/data>), retrieved July 2026. No data ship with the paper. The figure script in the repository recomputes each headline estimate from the public files and asserts it against the numbers of record before drawing.

Series	Outcome	Cutoff	Observations
GPQA-Diamond (Benchmarking Hub)	logit of mean score per model	o1-preview, 2024-09-12	180 models, 45 before
METR 50% time horizon	$\log_2$ minutes per model	o1-preview, 2024-09-12	48 models, 12 before
Epoch Capabilities Index	index points per model	o1-preview, 2024-09-12	455 models
AI chip owners	ln cumulative H100-equivalents, quarterly, vs US hyperscalers	export controls, 2023-10-17	15 quarters per group

Table 1: The four series. The logit removes the mechanical slope compression of a bounded score near its ceiling. The incomplete 2026-03-31 quarter is dropped from the chip series because its cumulative totals fall.

3 Methods

3.1 Estimator

Let x be days since the cutoff and $m[s]$ the one-sided derivative of $E[Y \mid x]$ at zero on side s . The kink is

$$\kappa = m[\text{right}] - m[\text{left}], \quad (1)$$

estimated by local linear fits with a separate intercept and slope on each side, triangular kernel weights inside the bandwidth, and HC1 standard errors with the degrees-of-freedom correction applied jointly across cells. The group DiK contrasts two such kinks at one date,

$$\tau = \kappa_{\text{China}} - \kappa_{\text{hyperscalers}}, \quad (2)$$

and identifies a policy effect only if the two groups' non-policy slope changes would have been parallel. All runs use a unit policy denominator, so κ is a descriptive slope change in outcome units per day.

Series	bw (d)	κ or τ / day	nominal z	placebos	nominal rej.	calibrated p	verdict
GPQA-Diamond logit	540	+0.00258	3.32	49	14%	0.020	dated bend
METR $\log_2$ 50% horizon	720	+0.00601	2.13	17	24%	0.278	era bend, not da
ECI, all models	540	−0.00725	−0.81	49	55%	0.780	null
China legal stock, DiK	548	−0.00154	−3.30	7	–	0.125 (floor)	inconclusive

Table 2: Headline specifications, triangular kernel, HC1 standard errors. “placebos” is the number of evaluable placebo cutoffs; “nominal rej.” the share of them at which HC1 rejects at 5%, a size diagnostic. The calibrated p is the add-one rank of the declared date among the placebos. The China row has seven evaluable positions, so its floor is 0.125.

3.2 Inference of record

For each declared cutoff the same estimator is run at shifted placebo cutoffs: the distinct running values in the central 80% of the support, at least a quarter bandwidth away from the declared date, thinned to at most 49 positions. The placebo-calibrated p is the add-one rank of the declared date’s $|z|$ among the placebo $|z|$ values. With N placebos it cannot fall below $1/(N+1)$, so at least 19 positions are needed for a 5% claim. A series with fewer positions is reported as inconclusive, whatever its nominal statistic.

The share of placebo dates at which the nominal HC1 test rejects is reported as a diagnostic of that test’s size on the series. It says nothing about power at the declared date and is never read as evidence for a null.

4 Results

Table 2 gives the headline specification for each series. Appendix Table 4 gives the bandwidth grid.

4.1 GPQA-Diamond at o1-preview

The logit-score slope rises from 0.00188 to 0.00446 per day at the release, about 15 to 34 raw percentage points a year (Figure 1). The declared date is the most extreme of 49 placebo dates at bandwidth 540 and 365 days, and of 46 at 730 days (calibrated p 0.020, 0.020 and 0.021). The largest placebo statistic at bandwidth 540 is $|z| = 2.78$, at 142 days before the release. The estimate is positive in all eight bandwidth-by-donut cells, a release-density kink is null ($t = 0.76$), and a training-compute covariate kink is null ($p = 0.54$).

The mechanism is composition. Reasoning models entered the release stream after September 2024, so the bend is a property of what was released, not of any model’s improvement.

4.2 METR time horizon at o1-preview

The $\log_2$ time-horizon slope rises from 0.00331 to 0.00932 per day at bandwidth 720, a doubling time of 9.9 months falling to 3.5 (Figure 2). The bend is real. It is not dated: a placebo cutoff 311 days before the release gives $|z| = 7.9$ against 2.1 at the declared date, and placebos at −196 and +455 days also exceed it. The calibrated p is 0.28 at bandwidth 720, 0.35 at 540 and 0.20 at 900. With 48 models the grid holds only 14 to 19 positions, so the floor is near 0.05 and

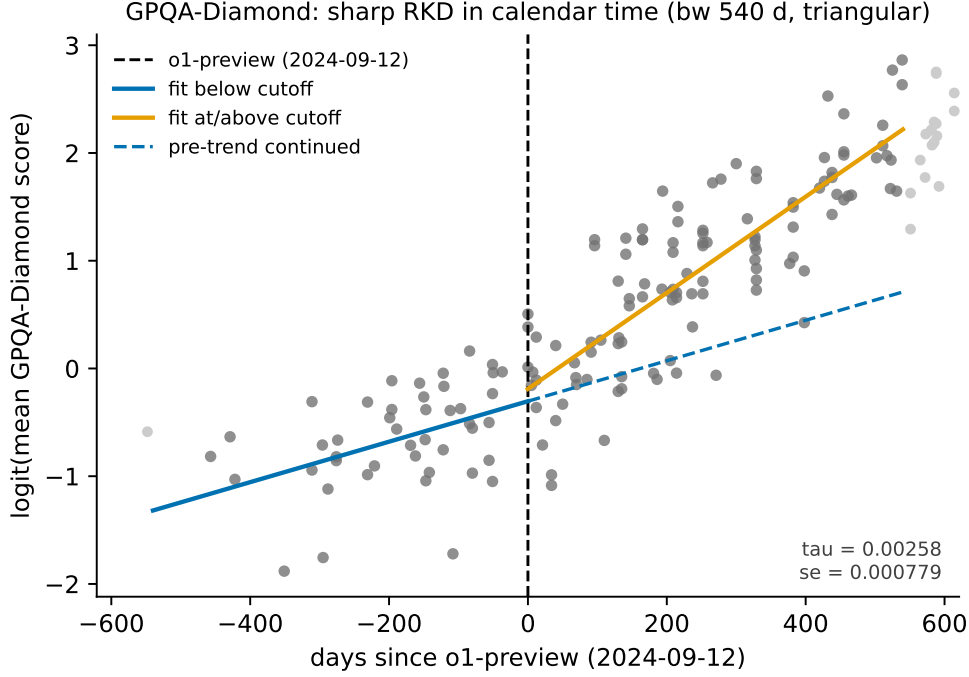


Figure 1: GPQA-Diamond, logit of mean score against days since o1-preview. Solid lines: local-linear fits on each side (bandwidth 540 days, triangular kernel); dashed: the pre-cutoff trend continued. The declared date is the most extreme of 49 placebo dates.

the test is close to its limit either way. The 80% horizon has three pre-cutoff models and is not estimable.

4.3 The Epoch Capabilities Index at o1-preview

The all-models index gains about 18 points a year on both sides of the release. The kink is -0.00725 points per day (se 0.00895) and null in every bandwidth cell (Figure 3). Its calibrated p is 0.78: 38 of 49 placebo dates give a larger statistic. The nominal test rejects at 55% of placebo dates on this series. The July version of this paper read those rejections as proof that the test had power. They show that nominal inference is oversized here, and nothing about power at the release. The null stands on its own.

4.4 China's chip stock at the October 2023 export controls

The export-licence schedule bent for China only, so the design is a group DiK: China's legal H100-equivalent stock against the four US hyperscalers at a common date, with the hyperscalers absorbing the global supply bend (Figure 4). The DiK is -0.00154 log units per day at bandwidth 548, about -0.56 per year, and -0.00114 at 730 days. The sign is negative in all 25 specifications of the July pass. The placebo-treated group, other buyers against the hyperscalers, is null ($t = 1.34$), and the October 2022 round, whose compliance loophole predicts no bite, is null ($t = 1.46$). The total stock including smuggled chips gives half the effect (-0.00076 , $t = -1.54$).

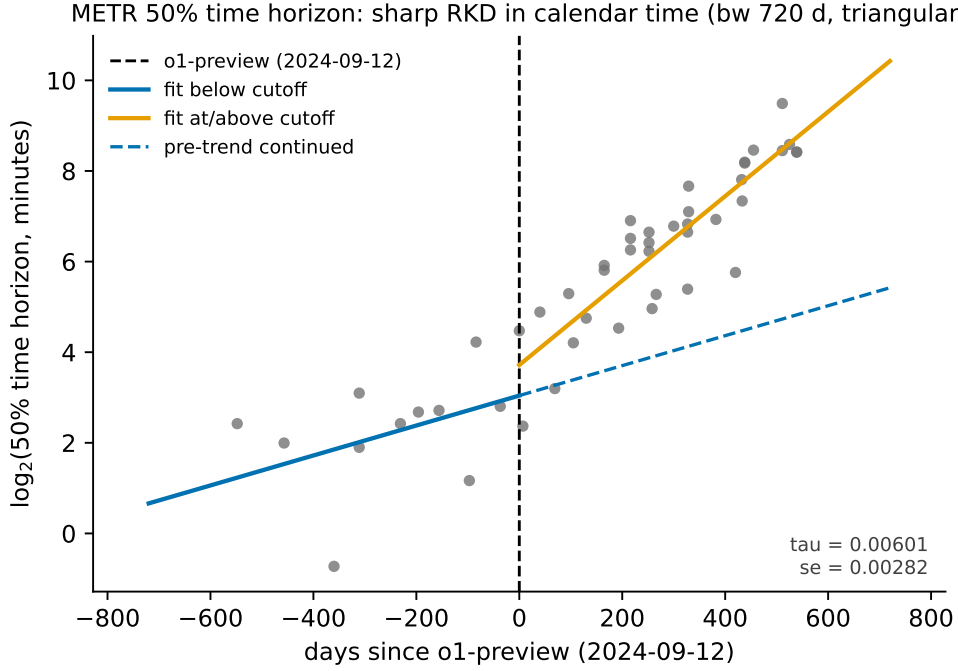


Figure 2: METR 50% time horizon ($\log_2$ minutes) against days since o1-preview. The slope nearly triples, but a placebo cutoff ten months earlier produces a larger statistic: the series bends over the era, not at the release.

None of this can be turned into a calibrated 5% claim. Fifteen quarters per group admit ten placebo positions, of which seven are evaluable, so the floor is 0.125 and the declared date reaches exactly that floor at both bandwidths. The design is inconclusive on this series. Table 3 records what quarterly data can and cannot show. The cells also hold six points each, where HC1 standard errors are known to be overstated [4].

5 Rejected designs

Three further claims were tested and do not survive.

Datacenter growth at ChatGPT. The GPU-cluster series (ln cumulative H100-equivalents by first operational date) has a nominal kink at ChatGPT’s release with $z = 9$ at the default bandwidth and $t = 12.9$ in the July pass. The nominal test rejects at 88% of placebo dates and the calibrated p is 0.62. A smoothly super-exponential cumulative series bends everywhere; the estimate is curvature.

Chinchilla. Tokens per parameter among language base models at the Chinchilla paper’s release (2022-03-29): $t = -0.91, -0.05$ and $+1.48$ at bandwidths 365, 540 and 730. The adoption entered the record as a ramp over two to three months, not a kink at the date.

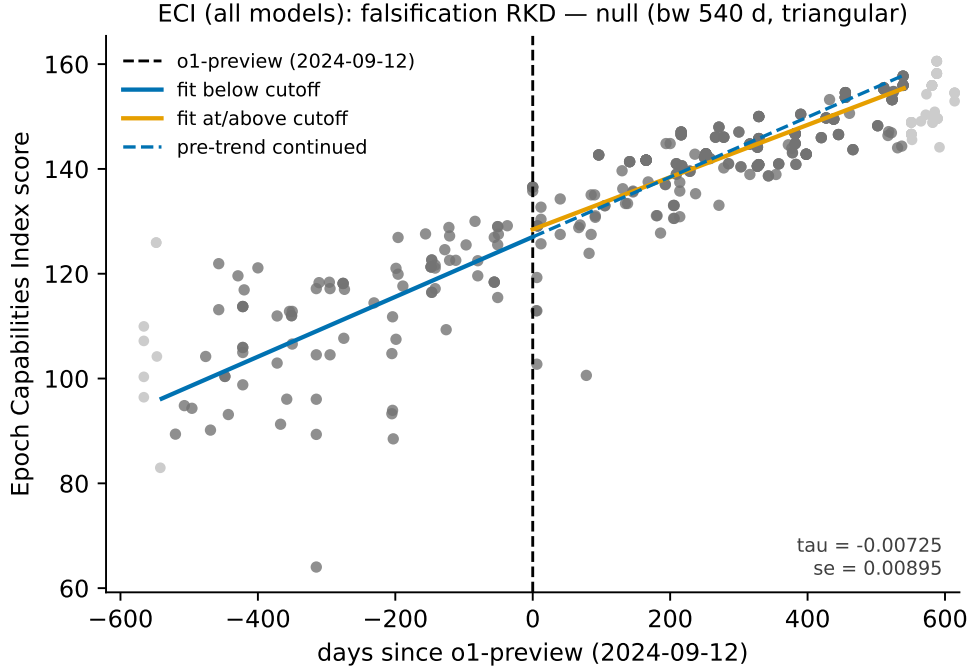


Figure 3: Epoch Capabilities Index (all models) against days since o1-preview. The two-sided fit and the continued pre-trend nearly coincide.

The EU AI Act threshold. The 10^{25} -FLOP line is a level discontinuity in obligations, not a slope change, so a kink is the wrong estimand. The running variable also sorts at the line (Fisher odds ratio 0.17, $p = 0.007$, for the post-Act deficit just above it), which rules out any regression-discontinuity reading. The bunching itself is analysed in a companion note.

6 Discussion

Three points carry beyond these series.

First, nominal robust inference is not usable for dating a bend in a short, serially dependent aggregate. On the four headline series it rejects at between 14% and 55% of dates where nothing happened, and at 88% on the cumulative datacenter series. Every verdict here rests on the placebo rank instead.

Second, placebo rejections are a size diagnostic, not a power certificate. The July version of this paper treated the ECI's rejections at shifted dates as evidence that its null at the release was well powered. That reading was wrong, and the correction changes no number but changes what the null means.

Third, the number of placebo positions is a property of the series, and it bounds what any calibrated test can say. Monthly or per-model series give 46 to 49 positions and a floor of 0.02. Fifteen quarters give seven positions and a floor of 0.125. A sign that is stable across 25 specifications on quarterly data is worth reporting, but it is not a dated effect at the 5% level and should not be described as one.

Contrast (vs hyperscalers)	bw (d)	τ / day	t	evaluable placebos	calibrated p
China legal stock	548	-0.00154	-3.30	7	0.125 (floor)
China legal stock	730	-0.00114	-2.85	6	0.143 (floor)
China total incl. smuggled	548	-0.00076	-1.54	7	0.125
China total incl. smuggled	730	-0.00046	-1.10	6	0.429
Other buyers (placebo group)	548	+0.00050	1.34	7	0.750
China legal vs neoclouds	548	-0.00094	-1.70	7	0.375
China legal, Oct 2022 round	548	+0.00182	1.46	—	—

Table 3: Group difference-in-kinks on quarterly ln cumulative H100-equivalent stocks. Evaluable placebo positions never exceed seven, so no row can reach $p < 0.125$.

Limitations. Calendar-time identification rests on the untestable assumption that no other event bent the series at the date; placebo grids rank the date, they do not certify the assumption. The METR grid holds 14 to 19 positions and sits near its floor. The GPQA bend is a composition effect of which models were released, and benchmark contamination drifts over time. The chip series are partly estimates, have six points per cell, and carry serial dependence that the estimator flags. All results describe the public record of AI progress, not any individual system.

Reproducibility

Point estimates come from the `natex` v0.2.0 kink module on the July 2026 extracts; placebo grids and calibrated p -values from `natex` v0.3.0 on the same extracts. The per-design records and the numbers of record live in the repository under `docs/case_studies/epoch-kinks.md`. The four figures regenerate from the public files via `paper/figures/make_figures.py`, which asserts every point estimate against that file before drawing.

Appendix: bandwidth grid

Series	bw (d)	κ / day	nominal z	placebos	nominal rej.	calibrated p
GPQA-Diamond logit	365	+0.00313	2.96	49	0%	0.020
GPQA-Diamond logit	540	+0.00258	3.32	49	14%	0.020
GPQA-Diamond logit	730	+0.00261	4.05	46	4%	0.021
METR log ₂ 50% horizon	540	+0.00507	1.41	19	16%	0.350
METR log ₂ 50% horizon	720	+0.00601	2.13	17	24%	0.278
METR log ₂ 50% horizon	900	+0.00627	2.37	14	29%	0.200
ECI, all models	365	-0.01190	-0.72	49	39%	0.660
ECI, all models	540	-0.00725	-0.81	49	55%	0.780
ECI, all models	730	-0.00463	-0.70	49	67%	0.820
GPU clusters at ChatGPT	708	+0.00080	8.95	49	88%	0.620

Table 4: Every bandwidth run for the per-model and monthly series, triangular kernel. Placebo grids and calibrated p -values computed with `natex` v0.3.0 on 2026-10-05.

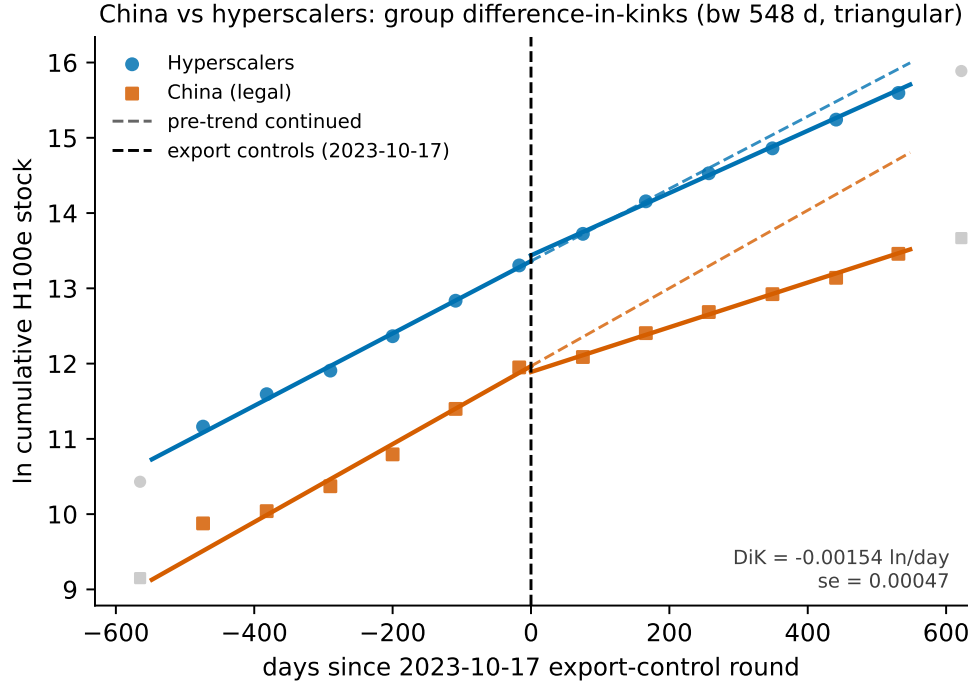


Figure 4: Group difference-in-kinks: $\ln$ cumulative H100-equivalent stock for China (legal) and the hyperscaler controls against days since the 2023-10-17 export-control round (bandwidth 548 days). Both series bend at the cutoff; the DiK is their difference.

References

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